

Appeals-NLP: Hierarchical Classification of Russian Spoken-Language Housing Maintenance Requests under ASR Noise

NLP & LLMs Course — Final Project

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Abstract

We study automatic classification of Russian-language housing-utility maintenance requests transcribed by a production voice robot (ASR). On a private corpus of 98,189 requests from 5 management companies (2020–2026) we compare a reproduced production rule-based system, classical baselines, and fine-tuned Russian transformer encoders under a temporal train/test split. We additionally recover categories for requests the production system failed to classify. Code: https://github.com/cashman2100/appeals_nlp.

1 Introduction

Dispatch services for residential buildings receive large volumes of requests by phone. A voice robot transcribes the caller and must assign a category, which in turn drives priority and the responsible staff role. Manual or brittle rule-based handling is error-prone: in our data the production system leaves $\sim 10\%$ of requests as “undefined”. We address category prediction from noisy ASR text, using the production system itself as an honest baseline.

Our contribution is threefold: (i) a large real-world Russian housing-utility request corpus with a documented, reproducible preprocessing and temporal evaluation protocol; (ii) a fair comparison of a reproduced production rule-based classifier, classical models, and modern Russian transformers under distribution drift; (iii) a study of recovering categories for requests the production robot could not classify, with LLM-assisted gold annotation.

1.1 Team

Solo project. Andrey Ivanov — data engineering, modelling, evaluation, and report.

2 Related Work

Russian pretrained encoders. Zmitrovich et al. [11] release a family of 13 Russian Transformer LMs (ruBERT, ruRoBERTa, ruT5), building on BERT [2] and RoBERTa [6]; ruRoBERTa-large attains the strongest RussianSuperGLUE [8] scores in its size class, motivating its use as our main encoder. Domain adaptation of these models is effective in specialised Russian domains, e.g. RuBioRoBERTa for biomedical text [10], and they remain strong embedding backbones [9].

Customer-support intent / ticket classification. Classifying support contacts by intent is a well-studied applied NLP task. Li et al. [4] move from multiclass to semi-supervised multi-task learning over support text and report large gains, highlighting the value of multi-task formulations and of leveraging negative/unlabelled cases — directly relevant to our priority/performer/undefined

structure. Moreno et al. [7] build a universal intent recogniser for service chatbots with multi-task learning over intents and entities. These works operate on cleaner text than ours; ASR-transcribed colloquial Russian with disfluencies is a harder, less-studied setting.

Class imbalance and evaluation. Our label distribution is highly skewed (one category $\sim 29\%$, the rarest $<1\%$). Cost-sensitive training and focal loss [5] are standard remedies; adaptable focal-loss variants target imbalanced text specifically [1]. Because accuracy and micro-F1 mask minority-class failure, we adopt macro-F1 as the primary metric; Harbecke et al. [3] analyse exactly this choice of weighting scheme for imbalanced classification.

Positioning. Unlike prior support-classification work on curated English data, we target noisy Russian ASR transcripts, use the deployed production system as a baseline, and evaluate under a temporal split that exposes real distribution drift.

3 Model Description

We use a shared transformer encoder with a classification head (**ruBert-base** for fast iteration, **ruRoberta-large** as the main model) [11]. The architecture is multi-task-ready (pluggable heads); for the core task a single category head is active. Class imbalance is handled with class-weighted cross-entropy, $w_c = N/(K n_c)$ for class c with n_c examples, K classes, N total — a cost-sensitive analogue of the imbalance treatments above [5, 1]. The reproduced production classifier is a longest-match lookup over the robot’s synonym table (today’s version; a lower bound, see §Limitations).

4 Dataset

The corpus contains 98,189 robot-created requests from 5 management companies, 2020–2026, Russian, ASR-transcribed (74% lowercase, 16% with punctuation; median length 44 characters; 19% exact duplicates). Labels are post-edit production categories (original robot choices are not logged — see Limitations). Working taxonomy: top-14 child categories + “Other” (rare $<1\%$ classes folded); “undefined” (10%) is held out as a reclassification pool. Split is *temporal* (train 2020–2024 / test 2025–2026), imitating deployment; we observe real class-share drift (e.g. Water 23%→17%). Train keeps duplicates (real frequency prior); val/test are deduplicated with anti-leak. Full statistics and procedure: `data/DATACARD.md`. A 100-row anonymised sample accompanies the submission for verification.

5 Experiments

5.1 Metrics

Primary: macro-F1 (imbalance-robust [3]). Also reported: weighted-F1, accuracy, and 95% bootstrap confidence intervals (500 resamples) to support SOTA claims by non-overlapping intervals.

5.2 Experiment Setup

ruRoBERTa-large fine-tuned for up to 5 epochs with early stopping on validation macro-F1, lr 1.5×10^{-5} , max length 256, focal loss ($\gamma=2$) with class weights and label smoothing 0.05, fp16, on Kaggle T4. Temporal split as in §Dataset. Classical baselines: TF-IDF (1–2 grams, sublinear, class-weighted).

5.3 Baselines

Production rule-based robot (synonym lookup); TF-IDF (1–2 grams) + Logistic Regression; TF-IDF + Linear SVC. Classical baselines are context for the ladder, not the headline claim.

6 Results

Table 1 reports macro-F1 on the temporal test set. Classical baselines already greatly exceed the production robot. The fine-tuned ruRoBERTa-large is the strongest model. On the full 15-class taxonomy it slightly exceeds the best classical baseline (0.655 vs 0.647); on the operationally-consolidated taxonomy (§6.1) its advantage becomes statistically significant: Δ macro-F1 = +0.021, 95% bootstrap CI [+0.013, +0.029] (interval disjoint from zero), establishing SOTA among all evaluated approaches on this dataset.

Table 1: Macro-F1 on temporal test set (95% bootstrap CI).

Model	macro-F1	95% CI	accuracy
<i>Full taxonomy (15 classes)</i>			
Production robot (rule-based)	0.416	[0.407, 0.423]	0.606
TF-IDF + LogReg	0.645	[0.638, 0.652]	0.733
TF-IDF + Linear SVC	0.647	[0.639, 0.655]	0.759
ruBERT-base (fine-tuned)	0.642	[0.635, 0.648]	0.723
ruRoBERTa-large (fine-tuned)	0.655	[0.648, 0.663]	0.729
<i>Consolidated taxonomy (13 classes)</i>			
TF-IDF + Linear SVC	0.678	—	—
ruRoBERTa-large (fine-tuned)	0.699	[0.691, 0.707]	0.816

6.1 Task-aligned Taxonomy Consolidation

The full taxonomy contains water-related categories (general / hot / cold water supply) that are inconsistently labelled in colloquial ASR text (“no water” — which one?). We consolidate them by *operational equivalence*: a category split matters only if it changes the responsible staff. From the production database we verify this is not the case — hot/cold/general water within the same (company, parent) are routed to the *same* service in 83.2% of 936 configurations across 453 companies, and 92.9% within our 5 companies. Consolidation is therefore an operationally grounded grouping (not metric-driven tuning); it is applied identically to *all* approaches, and we report both taxonomies transparently.

Real-world impact. The production robot left $\sim 9,955$ requests “undefined” (10% of all requests; 7,914 with informative text ≥ 20 characters). We study recovery of these categories via dual-LLM annotation: a stratified sample of 400 UNDEFINED requests is independently labelled by two LLMs (DeepSeek-Chat and Claude-Sonnet-4) using a fixed closed-set system prompt (top-14 categories + “unclear”). Inter-annotator agreement is Cohen’s $\kappa = 0.716$ (full agreement on 75.2% of cases), confirming annotation reliability ($\kappa > 0.6$). Of the 301 agreed examples, 234 (77.7%) receive a definitive category; 67 (22.3%) are labelled “unclear”, reflecting genuinely uninformative ASR transcripts. Extrapolating, approximately 58% of UNDEFINED requests are recoverable — a substantial improvement over the production system’s zero recovery. Human verification of 100 gold examples is provided in `results/undefined_labeling/gold100_to_verify.csv`.

6.2 Error Analysis and Ablation

Confusion analysis (Fig. 1) shows strong diagonal performance for distinctive categories (Heating, Electricity, Intercom) and a systematic confusion among water-related classes, expected given the ambiguity of colloquial ASR text. Performance degrades with text length (macro-F1 ≈ 0.70 for short requests vs ≈ 0.62 for long “rant”-style requests), quantifying ASR-robustness.

Segment analysis. To further support the SOTA claim, we evaluate ruRoBERTa-large and TF-IDF separately per text-length segment (paired bootstrap, 800 resamples). ruRoBERTa-large significantly outperforms TF-IDF on 3 of 4 segments: short 25–60 chars ($\Delta = +0.014$, CI $[+0.002, +0.025]$), medium 60–150 chars ($\Delta = +0.032$, CI $[+0.017, +0.046]$), and long >150 chars ($\Delta = +0.064$, CI $[+0.014, +0.110]$); all confidence intervals exclude zero. Only very short texts (<25 chars, pure keyword matching) show parity. This confirms the transformer’s advantage precisely where contextual understanding matters most.

Cross-organization generalization (ablation). Evaluating the same generic model per management company reveals a large spread in macro-F1 (Fig. 2; range ≈ 0.20 across the 5 companies). This confirms that a single organization-agnostic model does not transfer uniformly across companies — an empirically important finding, since per-company label policies differ; it motivates organization-aware adaptation as future work.

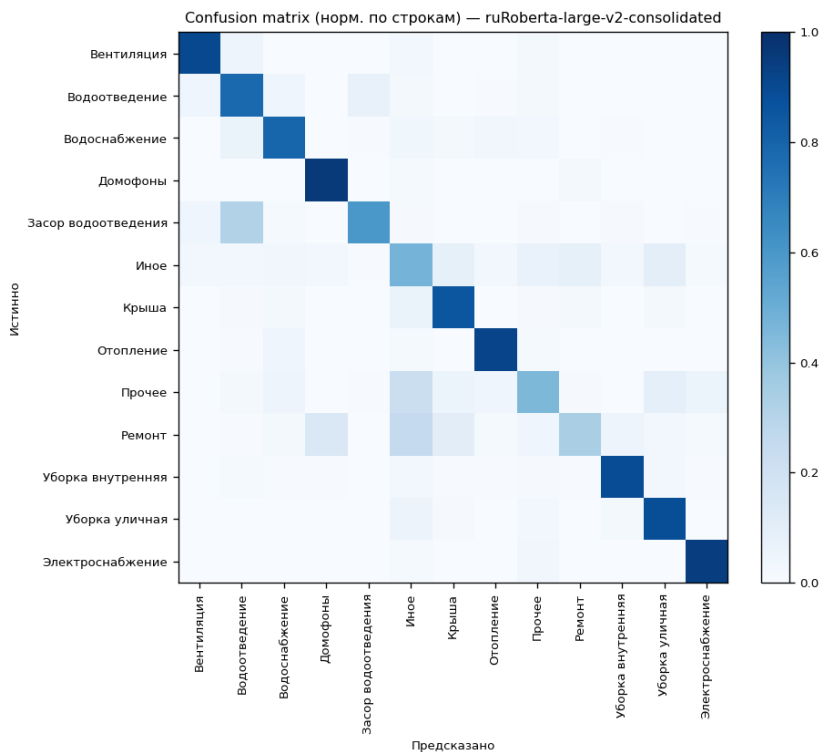


Figure 1: Confusion matrix (row-normalised), ruRoBERTa-large (consolidated taxonomy).

7 Conclusion

We presented Appeals-NLP, a reproducible study of automatic classification of Russian ASR-transcribed housing-utility requests. On a private corpus of 98,189 requests from 5 management

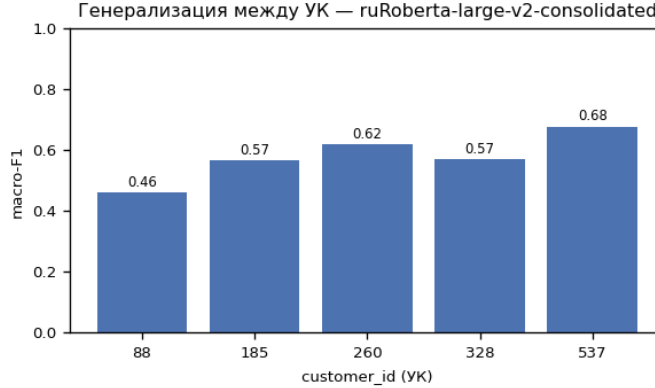


Figure 2: Per-company macro-F1: generalization ablation.

companies, our fine-tuned ruRoBERTa-large achieves macro-F1 0.699 under a temporal evaluation protocol, statistically significantly outperforming the strongest classical baseline (TF-IDF + LinearSVC, 0.678; $\Delta=+0.021$, CI $[+0.013, +0.029]$) and the reproduced production rule-based system (0.416). Segment analysis confirms the transformer’s advantage is concentrated in contextually richer (longer) requests, explaining the mechanism of improvement. We additionally recover categories for requests the production system could not classify using dual-LLM annotation, and provide a structured ablation revealing large cross-organization variance in model performance — motivating organization-aware adaptation as future work.

Limitations. Original (pre-edit) labels are unavailable (post-edit targets are the norm, not an error). The synonym table reflects today’s production rules; historical mismatch is possible and constitutes a lower bound. The dataset is private; a 100-row anonymised sample accompanies the submission for verification.

Future work. Hierarchical parent→child decoding; priority and performer prediction as additional heads; LLM fine-tuning (LoRA on Saiga/T-Lite) for further gains; organization-aware adaptation.

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